MILITARY SPECIFICATION
INFLATION-LAUNCHING DEVICE, BALLOON, METEOROLOGICAL ML-594( )/U

1 SCOPE

1.1 This specification covers one type of Launching Device for meteorological sounding balloons, which is capable of use with field type hydrogen generation equipment, designated Inflation Launching Device, Balloon Meteorological ML-594( )/U

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Fiberboard
PPP-B-640	Box Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiberboard, corrugated
PPP-S-760	Strapping, Nonmetallic, and connectors
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communication, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant - For the Treatment of Communications, Electronic, and Associated Electrical Equipment

FSC 6660

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SPECIFICATIONS (con't)

MIL-E-5272	Environmental Test, Aeronautical and Associated Equipment, General Specification for
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-hardening Steels or Alloys, Nickel Alloys, Heat-resisting Alloys and Titanium, Spot and Seam.
MIL-M-10578	Metal Conditioner and Rust Remover
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43291	Box, Fiberboard, Corrugated, Double-wall, Weather Resistant

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme Temperature Cycle
MIL-STD-170	Moisture Resistant Test Cycle for Ground Signal Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-2000500	Inflation-Launching Device, Balloon, Meteorological ML-594( )/U
SC-DL-2000502	Case, Inflation-Launching Device

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC-76-1964	Electrical Resistivity of Fabrics

(Application for copies should be addressed to Executive Secretary, P. O. Box 886, Durham, N. C. 27702).

3. REQUIREMENTS

3.1 Preproduction samples. - The contractor shall furnish two (2) preproduction samples for approval, if required by the invitation for bids and contract.

3.2 Construction. - Inflation and Launching Device ML-594( )/U shall be constructed in accordance with SC-DL-2000500 and Case, Inflation Launching Device in accordance with SC-DL-2000502.

3.2 1 Model. - A model of the equipment will be loaned to the successful bidder as an aid in interpreting the drawings.

3.3 Parts, materials, and processes; general. - Parts, materials and processes shall be as specified in the equipment drawings. Ratings and derated values of parts and materials shall conform to requirement 18 of MIL-STD-454

3.4 Cleaning. -

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3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with requirement 9 of MIL-STD-454, or as specified in an applicable document.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Finish, protective. - The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.6 Marking. -

3.6.1 General. - Marking shall conform to MIL-M-13231 and the applicable equipment drawings.

3.6.2 Serial numbers. - All completed equipments shall have serial numbers.

3.7 Tropicalization of material. - Fungus-inert materials shall be used where practicable. Such materials are listed in requirement 4 of MIL-STD-454.

3.7.1 Fungus. - Equipment and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of fungus test in accordance with paragraph 4.12.

*Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.7.2 Moisture and fungus proof (MFP) treatment for equipment & assemblies. - An MFP coating in accordance with MIL-V-173 shall be applied as specified in MIL-T-152 to equipments, assemblies and parts which have been cleaned prior to coating to remove such contaminants as lubricating oils, mold release agents, sand, corrosion products, solder fluxes, fingerprints and dust.

3.8 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.8 1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

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3.8.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. (See 4.16) Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.8.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.9 Weight. - The complete equipment, packaged for transit, shall not exceed 150 pounds.

3.10 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.11 Safety Measures. - The equipment shall be grounded in accordance with the applicable equipment drawing for protection against static electricity. The canopy shall have a minimum surface resistivity of 9×10^9 ohms per square inch.

3.12 Service conditions. - The equipment shall meet the following service conditions-- Where a test is referenced, meeting the test shall be considered as compliance with this requirement:

3.12.1 Equipment nonoperating. (Storage and transit). - The equipment shall comply with the operational requirements of 3.12.3 after subjection to any of the following nonoperating conditions successively or in combinations likely to be encountered during world wide short-term storage and transit:

(a) Temperature. - Air temperature from minus 80°F for a period up to 3 days with no solar radiation to plus 155°F for up to 4 hours with no solar radiation.

(b) Relative humidity. - Relative humidity up to 97 to 100 percent from minus 65°F to plus 85°F with condensation from plus 33°F to plus 85°F. Relative humidity equivalent to 13 grains of moisture per cubic foot of air between plus 85°F and plus 125°F.

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3.12.1 Equipment nonoperating. (Storage and transit) (Con't)

(c) Rain. - Two (2) inches of rainfall per hour, with wind at 50 mph ±5 mph for a period of 4 hours.

(d) Elevation. - Elevation up to 40,000 feet above sea level.

(e) Orientation. - Indefinite storage in any orientation.

(f) Dust and sand. - Dust and sand as encountered during desert storage and transit.

(g) Shock, drop. - Any equipment supplied in transit package shall be capable of withstanding the test of 4.14.

(h) Shock bounce. - The equipment as packaged for field use shall be capable of withstanding shocks induced in field use during transportation.

3.12.2 Equipment non-operating (standby) - The standby conditions are the same as above except as follows:

"(e), (g), and (h) of 3.12.1. These requirements do not apply."

"(i) Ice. - The equipment shall be subjected to a 1/4 inch radial ice loading while in the standby condition."

3.12.3 Equipment operating. - The equipment, as set up for operation, shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations.

(a) Operation canopy. - Continuous use for a period of 1 year without replacement.

(1) All other parts. - Continuous use for a period of 5 years with no more than normal maintenance.

(2) Compliance will be determined by proof of reliability.

(b) Temperature. - Ambient temperature in the range of +140°F to -40°F. (The 140°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, after the equipment has reached temperature equilibrium, and at the low temperature extreme not to exceed 24 hours, after the equipment has reached temperature equilibrium.

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3.12.3 Equipment operating. - (Con't)

(c) Elevation - Elevation up to 10,000 feet above sea level.

(d) Rain and wind. - The equipment, with an inflated or deflated balloon installed, shall withstand a 1/2 inch rainfall, per hour, with wind at 20 mph ±5 mph for 4 hours and gusts of wind up to 50 mph.

(e) Ice. - The device shall be subjected to freezing rain during the balloon installation, inflation, and launching cycles. Under this condition the total accumulated ice thickness shall not exceed 1/16 inch.

(f) Blowing dust. - Particles 0.0017 millimeter in diameter blowing at 50 mph.

3.13 Technical literature, tools, and running spare parts. - Technical literature, all tools required for installation, and operation, including special tools and running spare parts, shall be provided as specified in the bid request and contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.14 Workmanship. - The equipment shall be manufactured in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein:

(3.2) Construction

(3.4) Cleaning

(3.7) Welding

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows.

(a) Preproduction inspection (does not include preparation for delivery)

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4.2 Classification of inspection. - (Con't)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection:

(1) Quality conformance inspection before preparation for delivery.

(2) Quality conformance inspection of preparation for delivery.

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B and Group C (See Tables I, II, and III).

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary document as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1, through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - This inspection including sampling, shall conform to Table I and the inspection procedures of MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE I - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Anti-Static Test	3.11	4.6	1.0%	
Visual-Mechanical	3.14	4.16	1.0%	4.0%

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall normally be preformed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. The AQL shall be 4 percent defective and the inspection level shall be S-4 for normal, tightened, and reduced inspection.

4.5.2.2 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II - GROUP B INSPECTION

Inspection	Req Para	Insp Para
Bounce	3.12.1(h)	4.13
Shock-Drop	3.12.1(g)	4.14
Interchangeability	3.10	4.18

4.5.3 Group C inspection. - This inspection shall consist of the test specified in Table III and shall be performed on sample units that have been subjected to and met Groups A and B inspection, sample units shall be selected in accordance with paragraph 4.5.3.1.

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TABLE III - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Altitude	3.12.1(d) & 3.12.3(c)	4.15
Temperature	3.12.1(a) & 3.12.3(b)	4.7
Moisture-Resistance	3.12.1(b)	4.8
Rain and Wind	3.12.1(c) & 3.12.3(d)	4.9
Dust and Sand	3.12.1(f)	4.10
Blowing Dust	3.12.3(f)	4.10.1
Ice	3.12.2(i) & 3.12.3(e)	4.11
Fungus	3.7	4 12

4.5.3.1 Sampling for inspection.-

(a) Two units shall be selected at random from the first ten (10) units produced These units shall be subjected to all tests in Table III. When preproduction samples are required by the contract, the testing of two (2) units from the first ten production units will be omitted.

(b) In addition to (a) above, one (1) unit shall be selected at random from each fifty (50) units produced, or fraction thereof. This unit shall be subjected to the "temperature" and "moisture-resistance" tests listed in Table III.

4.5.3.2 Noncompliance.- The contractor shall report in writing each Group C failure occurance, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the preposed corrective action on (1) The process and materials, as applicable, and, (2) All units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted, however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that corrective action was effective.

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4.5.4 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Anti-static test - The balloon canopy shall be examined to determine the anti-static condition of the canopy material It shall be required to meet surface resistivity 9×10^9 ohms per square inch (minimum) when tested in accordance with AATCC specification 76-1964.

4.7 Temperature test. - The equipment shall be subjected to the temperature cycle shown on MIL-STD-169 as modified in 3.12.1(a) and 3.12.3(b)

4.8 Temperature-humidity test. - The equipment in a ready to operate condition shall be placed in a suitable humidity chamber and tested as follows.

4.8.1 Test conditions. -

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start tests not more than 5 minutes after humidity cycle has been completed.

4.8.2 Test methods. -

(a) Dry at 130° ±5°F for 24 hours

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Subject to continuous cycling for two 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.

(d) After cycling has been completed the equipment shall meet full specified performance.

4.9 Pain and wind test. - The equipment shall be tested, step by step, as follows

(a) Dry at 150° ±5°F for 48 hours.

(b) Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity

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4.9 Rain and wind test. - (Con't)

(c) Take measurements to insure that the equipment meets specified performance.

(d) Place the equipment as stored or in its normal operating position, in a test chamber maintained at 77° ±5°F.

(e) Subject the exposed surfaces of the equipment to simulated rainfall. The water shall be at a temperature of 65° ±5°F and shall fall freely at a rate of 2 inches per hour in the stored and standby conditions, and 1/2 inch per hour in operating conditions. Maintain this condition for 1 hour, during which time performance measurements shall be taken.

(f) With the equipment in a nonoperating condition and rainfall continuing at the required rate, introduce simulated wind blowing horizontally at 50 ±5 miles per hour against one side of the equipment. Maintain this condition for 1 hour, on each of the four sides.

(g) With the equipment in the operating condition and rainfall at the required rate, simulate a wind blowing horizontally at 20 ±5 miles per hour, with gusts up to 50 miles per hour. Maintain this condition for 1 hour, during which time performance measurements shall be taken. Repeat for each of the four sides. Equipment shall meet the requirements of paragraph 3.12.3(d).

4.10 Sand and dust test. - The equipment shall be tested as follows.

(a) Place the equipment in its transit case, in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space The relative humidity shall not exceed 30 percent at any time during the test.

(b) The sand and dust shall conform to the requirements of MIL-E-5272, paragraph 5.4.3 or 4.11.1(a) through 4.11.1(e).

(c) Maintain internal temperature of test chamber at 77° ±4°F for a period of 6 hours, with sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(d) After 6 hours at above conditions, raise the temperature to 160° ±4°F and maintain it at this value for 6 hours. The other conditions shall remain the same.

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4.10 Sand and dust test. - (Con't)

(e) At the completion of the above test cycle, take the equipment out of the chamber, allow to cool to room temperature, remove equipment from transit case and assemble for operation. With the equipment in the standby condition repeat paragraph (a) through (d).

(f) Remove equipment from test chamber and allow it to cool to room temperature. Accumulated dust shall be removed by brushing, wiping, or shaking, with care taken to avoid introduction of additional dust into the equipment However, dust shall not be removed by either blast or vacuum cleaning The equipment shall meet all specified performance requirements and the requirements of paragraph 3.12.1(f) There shall be no evidence of deterioration of either external or internal parts.

4.10.1 Blowing dust. - Demonstration of the equipment to function satisfactorly under the conditions of paragraph 3.12.3(f) will satisfy this requirement

4.11 Ice test. -

4.11.1 Equipment nonoperating (standby). - Demonstration of the ability of the equipment to operate satisfactorly after being subjected to the conditions of paragraph 3.12.2(i) will satisfy this requirement.

4.11.2 Equipment operating. - Demonstration of satisfactory operation will satisfy the requirements of paragraph 3.12.3(e).

4.12 Fungus - The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810 For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition.

4.13 Bounce test. - The equipment as packaged for transit, shall be tested on the package tester for military equipment of suitable capacity, as made by the L.A.B. Corporation, Skaneateles, New York, or equal The test shall be as follows·

(a) Cover the test bed of the package tester with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space six-penny nails, with heads below the surface, at 6 inch intervals around all four edges and at 3 inch intervals in a 6 inch square in the center.

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4.13 Bounce test. - (Con't)

(b) The equipment as secured in its transit, or combination case, or as otherwise prepared for field transportation shall be placed on the table of the package tester. Suitable wooden fences shall constrain the equipment from a horizontal motion in a direction parallel to the axis of the shafts of the package tester of more than 2 inches, and in a direction perpendicular to the axis of the shafts of the package tester of more than a distance sufficient to insure that the equipment will not rebound from fence to fence. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ±1 percent, for a total of three hours. At the end of each 1/2 hour period, the equipment shall be placed on a different face, so that at the completion of the test, the equipment has rested on each of its six faces.

(c) After completion of the test, the equipment shall meet full specified performance.

4.14 Shock test; drop. - The equipment as packaged for transit shall be subjected to a drop of 30 inches on each face, edge, and corner to a total of 26 drops. With its longest dimension parallel to the floor, the transit case, with the equipment within, shall be supported at the corner of one end by a block 5 inches in height, and at the other corner of the same end by a block 12 inches in height. The opposite end of the case shall then be raised to the specified height and allowed to fall freely. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

4.15 Altitude test. -

4.15.1 Test procedure (equipment operating). - Place the equipment in an altitude chamber in a standby condition with the balloon in the canopy ready for inflation. The balloon should be as free of air or gas as possible to prevent partial inflation during pressure reduction. If partial inflation does, occur, allow for it in operating the equipment at high altitude. Set up the equipment so that the gas can be controlled from outside the chamber. Reduce the pressure in the chamber to 20.58 ±.1 inches of mercury. Maintain this pressure for 2 hours for standby, then inflate the balloon

4 15.1.1 Proper inflation of the balloon shall be considered successful completion of the test.

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4.15.2 Test procedure (equipment nonoperating). - Place the equipment, in its transit case, in an altitude chamber. Reduce the pressure in the chamber to simulate an altitude of 40,000 ft. maintain this condition for 1 hour. After the completion of this test, return the chamber to ambient pressure, remove the equipment from the chamber and demonstrate that satisfactory operation may be achieved. Successful demonstration shall be considered as meeting the requirements of this test.

4.16 Visual and mechanical inspection. - The equipment shall be examined in accordance with paragraph 4.17 and major and minor defects as specified in MIL-STD-105.

4.17 Surface inspection. - Equipment shall be visually inspected for the following

(a) Workmanship, assembly and fit, mechanical safety and marking.

(b) Welding.

(c) Materials, parts, and finishes.

(d) Treatment for prevention of corrosion

(e) Securing of parts.

4.18 Inspection for interchangeability. - The dimensions shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a dimension is not within specified or design limits, it shall be considered a major defect.

4.19 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be as specified Level A or C (See 6.2(c)).

5.1.1 Level A

5.1.1.1 Cleaning. - All components of the ML-594( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. - All components of the ML-594( )/U shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical manual. - Each technical manual shall be packaged method 1C-1.

5.1.1.4.2 ML-594( )/U. - Each ML-594( )/U, shall be packaged method III, as follows: Place all the components of the ML-594( )/U within the transit case and secure all fastenings and ties. Close the zipper. Cushion the case on all surfaces with cells and pads or both, fabricated of fiberboard conforming to PPP-F-320, W6c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned case within a close-fitting fiberboard box conforming to PPP-B-636W5c. Place the technical manual packaged as specified in 5.1.1.4.1, in the box, directly under the lid. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - ML-594( )/U shall be packaged in a manner that afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C as specified (See 6.2(c)).

5.2.1 Level A. -

5.2.1.1 ML-594( )/U. - Package as specified in 5.1, shall be packed within fiberboard boxes conforming to PPP-B-640, class 2, grade A style E or MIL-B-43291. Gross weight of boxes conforming to PPP-B-640 shall not exceed 200 pounds. Box closure shall be in accordance with the appendix of the box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II respectively: selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. - ML-594( )/U packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that the boxes shall be domestic class conforming to PPP-B-636, DW, special requirements use, or PPP-B-640 class 1 grade A, style E. Containers, conforming to PPP-B-640, shall be reinforced as specified in 5.2.1, except nonmetallic strapping shall be type 1 conforming to PPP-S-760.

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5.2.3 Level C. - ML-594( )/U packaged as specified in 5.1, shall be packed within containers of the type size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Serves as an inflation shelter and launching platform for meteorological balloons at an altitude up to 10,000 feet in winds up to 50 mph, and at temperatures in the range of -40°F to +140°F.

6.2 Ordering Data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of preservation, packaging and packing (See section 5).

(d) If rough handling and functional tests are required.

(e) If preproduction samples are required.

(f) Place of final inspection.

(g) Technical literature required.

(h) Quantity of tools and running spare parts required.

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, ML-594W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

CUSTODIAN
ARMY-EL

PREPARING ACTIVITY
ARMY-EL
PROJECT NO. 6660-A139

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-I-55538(EL) INFLATION-LAUNCHING DEVICE, BALLOON, METEOROLOGICAL ML-594( )/U

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

___ DIRECT GOVERNMENT CONTRACT ___ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

___ YES ___ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

ESC FM 2011 66

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U. S. Army Electronics Command
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Fort Monmouth, New Jersey 07703

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